Decentralised Single-Index Bandits

Backup slides: proofs, extra experiments and earlier findings

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Course: CS6007: Multi-Agent Machine Learning

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The flood relay used in the early experiments

Phase 1: exact pooling without a server

Stein sums are **sufficient statistics**: spanning-tree aggregation (up and back down a BFS tree, $2\cdot$depth rounds) gives *every* agent the exact centralised $\hat{\theta}$.

Freeze: one shared grid

All agents freeze the same $\hat{\theta}$; the window W comes from max-consensus over D rounds.

Phase 2: event-triggered flooding of bin tables

Each agent stores the freshest copy of every agent's own (n_j, S_j) table, relayed one hop per round. An agent republishes only when its unpublished count in a bin exceeds $\gamma\times$ what it currently sees.

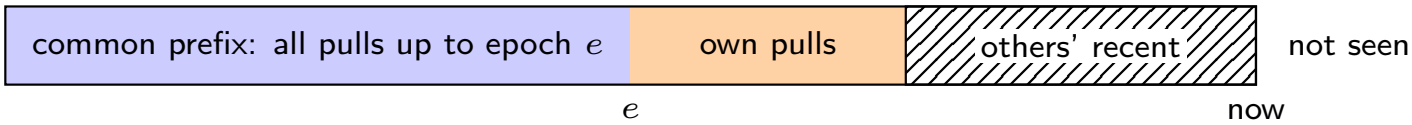
Design principle

Relay exact, sparse updates instead of averaging.

Our first proof: synchronised epochs

Phase 2, as analysed (Λ = delivery time, = D on a static graph)

- Every $L = 2\Lambda$ rounds, an agent raises an **alarm** if its own new pulls in some bin exceed $\gamma\times$ the network count at the last epoch. Any alarm $\Rightarrow$ an **epoch**.
- At an epoch every agent floods a **snapshot** of the bins it pulled since the last one.
- Every agent acts on the **same** epoch table $G(e)$ plus its **own** pulls since e.



Why this version

What an agent sees is *one common prefix plus one own block*, so the confidence width is exactly the server's. In the code version each agent sees N separate per-agent prefixes, and the proof loses a factor $\sqrt{N}$.

Theorem 17: regret with synchronised epochs

Synchronised epochs: a Phase 2 variant in which all agents act on the *same* delayed pooled table plus their own pulls (see backup). It was the first version we could analyse; Theorem 27 (main deck) now covers the relay we run.

Theorem (with probability $\geq 1 - 3\delta$)

$$R_T(N) \leq \underbrace{2L_f N(T_0+\Lambda)}_{\text{Phase 1}} + \underbrace{4\sqrt{2}\, c_\delta \sqrt{(1+N\gamma)N_b NT}}_{\text{UCB}} + \underbrace{2L_{f'}\Delta NT}_{\text{bin width}}$$
$$+ \underbrace{4L_{f'} X_{\max}\varepsilon_0 NT}_{\text{direction error}} + \underbrace{2L_f N_b\big(2+N(\gamma+4\Lambda)\big)}_{\text{delay: the only graph term}}$$

Corollary (rate): $\Delta = (NT)^{-1/3}$, $\gamma = \gamma_0/N$, T_0 balanced

$$R_T(N) = \tilde{O}\Big((d\,NT)^{2/3} \;+\; N\Lambda\,(NT)^{1/3}\Big)$$

Matches Theorem 5 in N and T when $N\Lambda \leq (NT)^{1/3}$: **the graph costs only an additive term.**

Proof in three steps

1. *Confidence:* view = common prefix + own block $\Rightarrow$ server width, no $\sqrt{N}$.
2. *Per pull:* regret $\leq 2\times$width + bin width (works for any sleeping pattern).
3. *Stale counts:* $\leq eG + 4\Lambda$ unseen pulls per agent per bin.

Theorem 22: the graph's cost does not grow with T

Observation

Theorem 17 lets *every* bin have $\approx 4N\Lambda$ unseen pulls (a factor $N_b \sim (NT)^{1/3}$). But a pull into bin j needs an arm in bin j, and there are only K arms per agent-round, so in total about $N\Lambda K$ pulls are unseen.

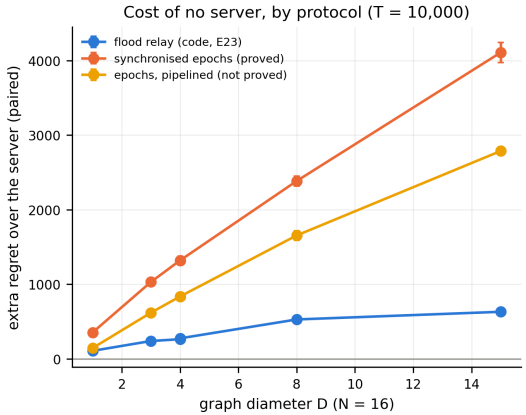
Theorem 22 (w.p. $\geq 1 - 4\delta$; Lemma 21: Chernoff on how often each bin is available)

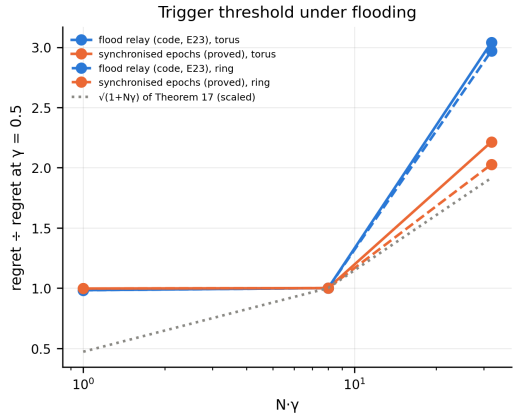
$$\text{delay} \leq \underbrace{(2L_f + 4\sqrt{2}c_\delta)16N\Lambda \min(K, N_b)}_{\text{graph; no } T,\ N_b} + \underbrace{\tilde{O}(N_b)}_{\text{no graph}} \ \Rightarrow\ R_T(N) = \tilde{O}\Big((dNT)^{2/3} + N\Lambda K\Big)$$

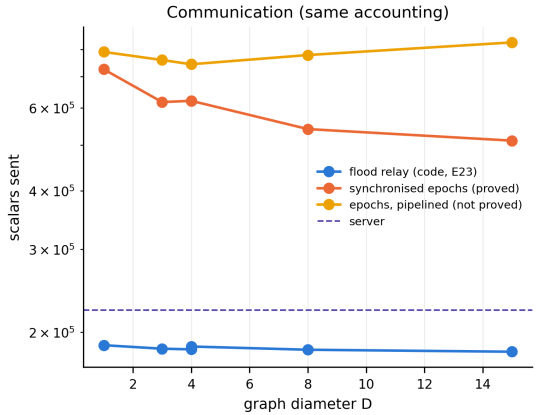
epochs	NDK	measured	ratio
complete	320	353	1.10
expander	960	1,030	1.07
torus	1,280	1,319	1.03
ring	2,560	2,384	0.93
path	4,800	4,107	0.86

- Shape NDK matches the measured excess within $\pm 14\%$ on every graph (no fitted constant; the proved constant is larger).
- Almost flat in T, as predicted.
- Theorem 19 ($\Lambda \to \sqrt{\Lambda}$) is better when K is large.

E31: the protocol we prove vs the protocol we run







- **Proved (synchronised epochs):** $+5\%$ ($D{=}1$) to $+59\%$ ($D{=}15$) over the server; grows like $D^{0.9}$.
- **Run (flood relay):** $+1.5\%$ to $+9\%$; grows like $D^{0.66}$.
- Pipelined epochs (not proved): a third of the gap.

- **Trigger:** $\gamma \le 0.5$ is free for both ($\le 2\%$); only $\gamma = 2$ costs ($\times 2$–3).
- **Scalars:** epochs send 2.4–$3.8\times$ more (every agent snapshots at every epoch).

Why: agreement costs latency

To use a common table, every agent waits for the *farthest* agent ($\approx 4D$ rounds). The relay uses nearby agents' pulls after a few hops. **Closed by Theorem 27:** the published-only relay is provable and performs like the relay (E36).

Mixing vs relaying: building a network sum

Every agent needs network totals: $\sum_i b_i$ (Phase 1) and n_j, S_j per bin (Phase 2). Toy example: ring of 4 agents with values $(4, 0, 0, 0)$; true average $= 1$.

Mixing (consensus / gossip)

Each round: $x_i \leftarrow \sum_j P_{ij} x_j$ (average with neighbours)

round	a_1	a_2	a_3	a_4
0	4	0	0	0
1	1.33	1.33	0	1.33
2	1.33	0.89	0.89	0.89
3	1.04	1.04	0.89	1.04

$\to 1$ but **never exact**; speed set by the *spectral gap*. Sends the whole state every round.

Relaying (flooding / message passing)

Each round: forward fresher copies of *each agent's own* record (with a version stamp).

round	who knows "a_1 has 4"
1	a_2, a_4
2	everyone

Exact after D rounds (diameter); sends only records that changed.

UCB uses counts directly $(1/\sqrt{n_j})$: mixing gives *approximate* counts, relaying gives *exact but stale* ones.

When does relaying win? Scope and limitations

Our claim (not "relaying beats mixing" in general)

For nonparametric bandits whose sufficient statistic is a **sparse table** (bin counts and sums), relaying exact, event-triggered updates **matches the regret** of consensus-based protocols with **orders of magnitude fewer scalars**, and is more robust to sparse or unreliable graphs.

Limitations of the flood relay (code)

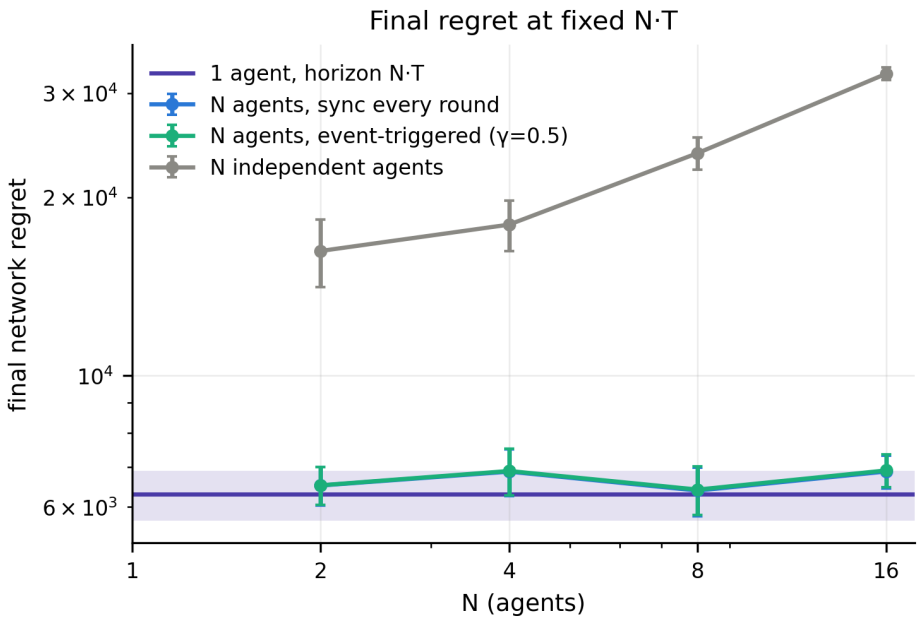
- Memory $O(N \cdot N_{\text{bins}})$ per agent
- Agents need unique IDs (no double counting)
- Individual agents' statistics are shared

What the epoch protocol fixes (Prop 30)

- **Memory** $O(N_{\text{bins}} + N)$**:** keep only the summed table
- **Tree version:** $O(N_{\text{bins}})$, no global IDs, only subtree partial sums are seen
- **Privacy:** Phase 1 provably private (Thm 35), but costly; Phase 2 open

Why most papers mix: constant memory in N, anonymous agents, built-in blending of data, a mature spectral-gap toolkit, and communication counted in *rounds*, where both cost one message per edge. **Precedent for relaying:** Dubey & Pentland (Coop-KernelUCB) use message passing for the same reason: averaging biases the estimates.

Reference: a server = one agent with NT pulls



E22: fixed budget $NT = 160$k, 20 paired trials (quadratic link)

- N synced agents: within **0–10%** of one agent at horizon NT, never better.
- Independent agents: **2.3–5.1**$\times$ worse.

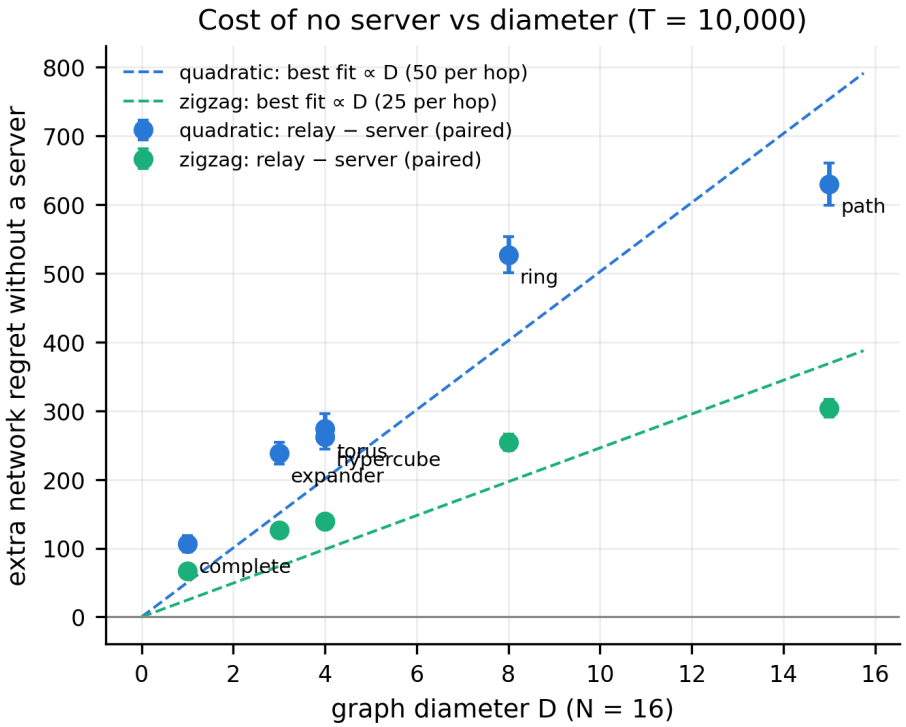
Proved (server)

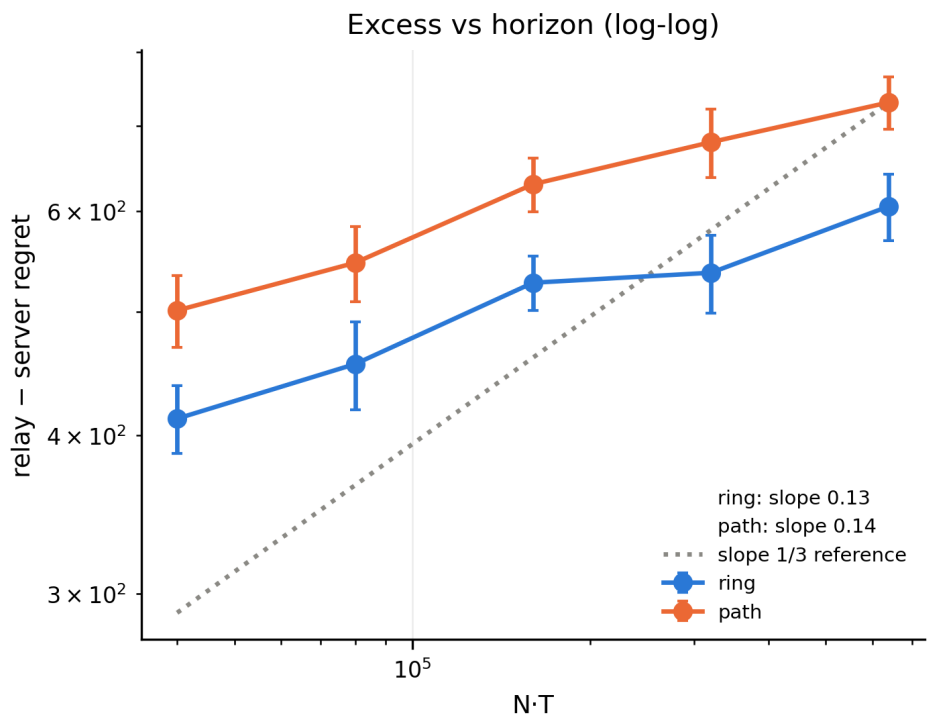
Lower bound by simulation: any N-agent algorithm $\geq c(NT)^{2/3}$.

Upper bound: Fed-ZoomSIB $= \tilde{O}\big((NT)^{2/3}\big)$.

The server rate is *closed*, so it is the yardstick for the serverless version.

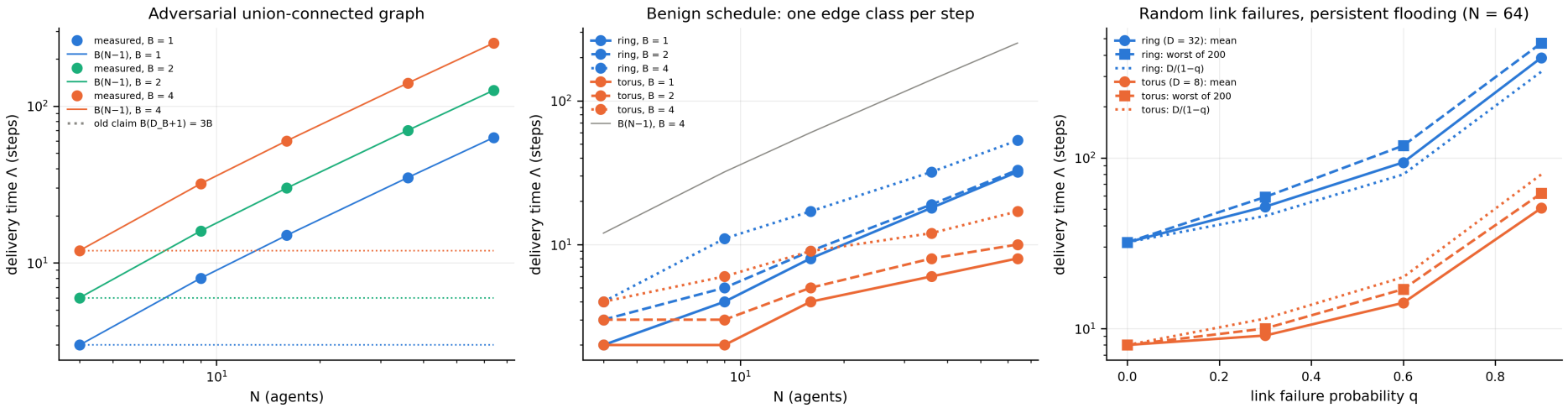
Finding 3: what losing the server costs (E23)





- Extra regret: **+1.5%** (complete, $D = 1$) to **+9%** (path, $D = 15$), $N = 16$; grows *sublinearly* in D.
- Nearly flat in the horizon: log-log slope **0.13–0.14** against the $1/3$ our bound predicts.
- Relay sends 15–20% *fewer* scalars than the server (no broadcast to all N).

E32: delivery time, measured



- **Adversarial star:** exactly $B(N-1)$ in every case; the old claim $3B$ is off by up to $21\times$ at $N = 64$.
- **Benign schedule** (one edge class per round): $\Lambda \leq B \cdot D$, far below the worst case.
- **Link failures:** worst of 200 runs within $1.5\times$ of $D/(1-q)$ (ring and torus, $N = 64$).

Proposition 34: stopping Phase 1 without a server

Proposition 34 (serverless stopping with agreement)

At each fixed checkpoint every agent floods its local Stein sum. Λ rounds later, every agent applies the *server's* rule to the *same* pooled sums. So all agents stop at the server's checkpoint and freeze the same $\hat{\theta}$ ($\eta_{\text{agree}} = 0$). Cost: $\leq 2L_f N\Lambda$ regret and $2|E|N(d{+}1)$ scalars per checkpoint.

ring ($D = 8$), E33	stops at	disagree	÷ fixed
fixed $T_0 = 60$	60	0	1.00
each agent, "any"	24	0.058	1.67
each agent, "all"*	50	0.024	1.07
server rule	28	0	1.34
pooled (Prop 34)	36	0	1.45

* assumes detection is free. Tolerance 0.15; 0.05 gives the same picture.

- **Exactly as proved:** stops *exactly* D rounds after the server (4, 8, 15) in every one of 20 paired trials, with zero disagreement.
- **But adaptive stopping itself loses** to a fixed T_0, even with a server ($\times 1.3$–1.4). The drift rule is the weak point, not the graph.

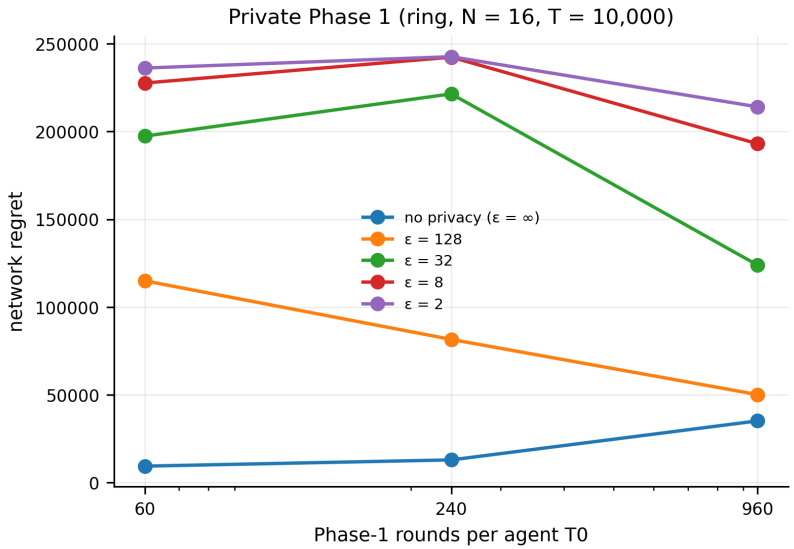
Theorem 35: a private Phase 1

Theorem 35

Each agent adds Gaussian noise once to its truncated Stein sum. Then everything the network does is (ε, δ)-DP for every Phase-1 sample, and the direction error grows by a factor $1 + \kappa$, with

$$\kappa = \sqrt{6dN \log(1.25/\delta)}/\varepsilon.$$

A longer Phase 1 buys it back: the leading term becomes $\tilde{O}\left(((1+\kappa)dNT)^{2/3}\right)$.



E35: the constants make it impractical at our scale

The noise scales with the worst-case truncation level ($L_f = 48$ for the quadratic link), about $20\times$ a typical summand. $\varepsilon \leq 32$ gives a random direction. $\varepsilon = 128$ costs $\times 12$ at $T_0 = 60$ and $\times 5$ at $T_0 = 960$: the trend the theorem predicts, but far from free.

Open: a smaller clipping level (bias vs noise); private *Phase 2*, where actions leak the agent's own rewards.

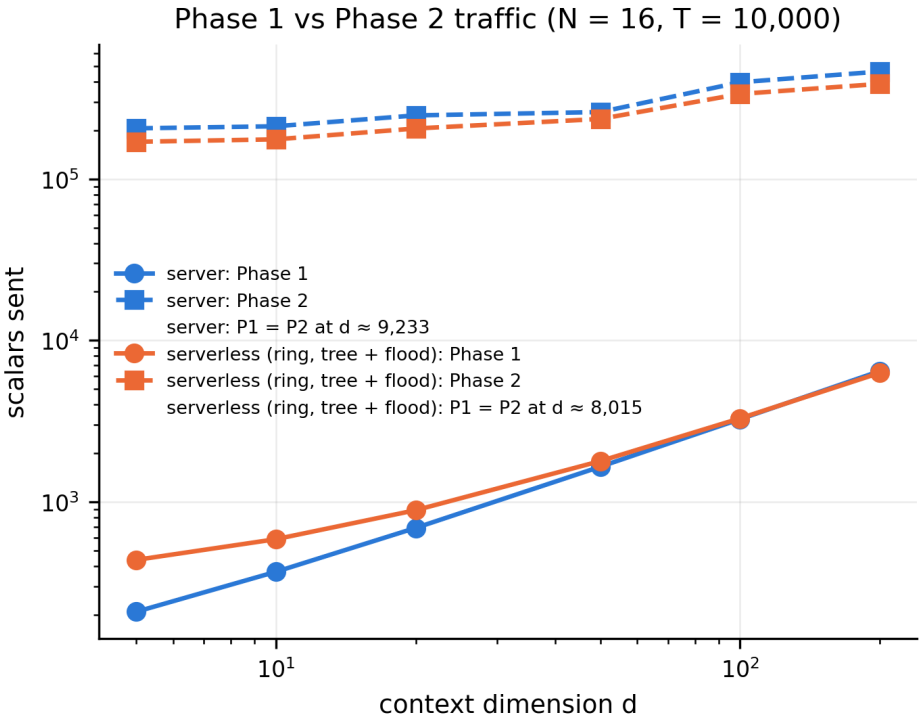
Checking the proofs: what changed

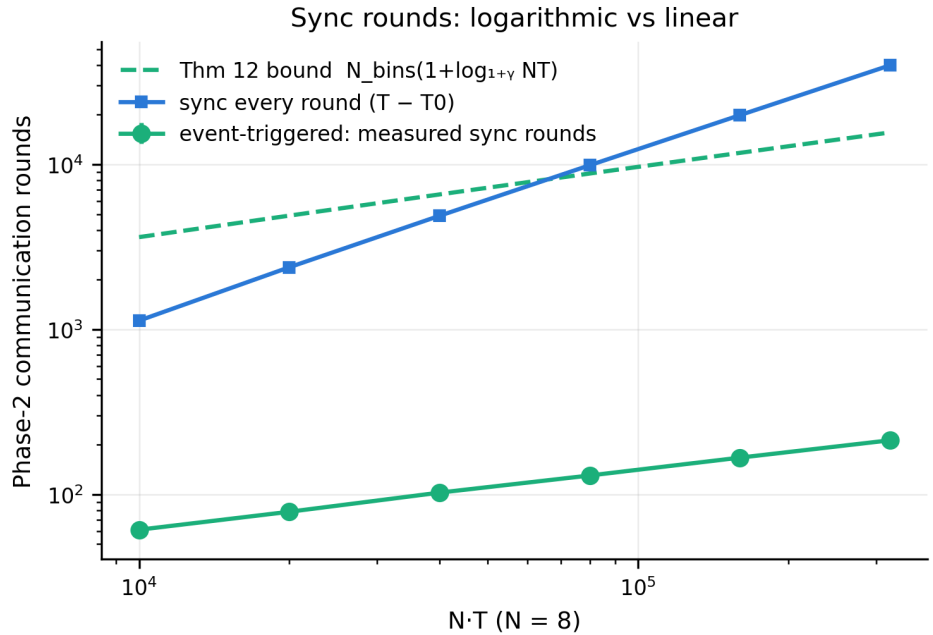
We re-checked every step against DBG's paper. Five changes:

1. **Noise:** DBG's Stein lemma (B.1) is proved for i.i.d. samples, but their assumption is only conditional sub-Gaussianity. We state i.i.d. noise explicitly.
2. **Hard instances:** DBG's lower bound uses $K \approx n^{1/3} \log n$ arms and $\mu^* \sim 1/\log n$, not a fixed K. Our Theorem 5 states this class.
3. **Expectation vs high probability:** f is bounded only near 0, so the regret theorem is stated with high probability (as DBG do). It holds in expectation when f is bounded, as on the hard instances.
4. **Time-varying graphs:** our first bound $B(D_B+1)$ was wrong. The star example breaks it, and the correct bound is $B(N-1)$, which is tight (E32).
5. **Simpler trigger:** comparing with the latest epoch the agents already hold halves the epoch count: $|\mathcal{E}| \leq 1 + N_b(1 + \log_{1+\gamma} NT)$.

Each point is logged with context in `docs/proofs/remarks.md` (R1–R11).

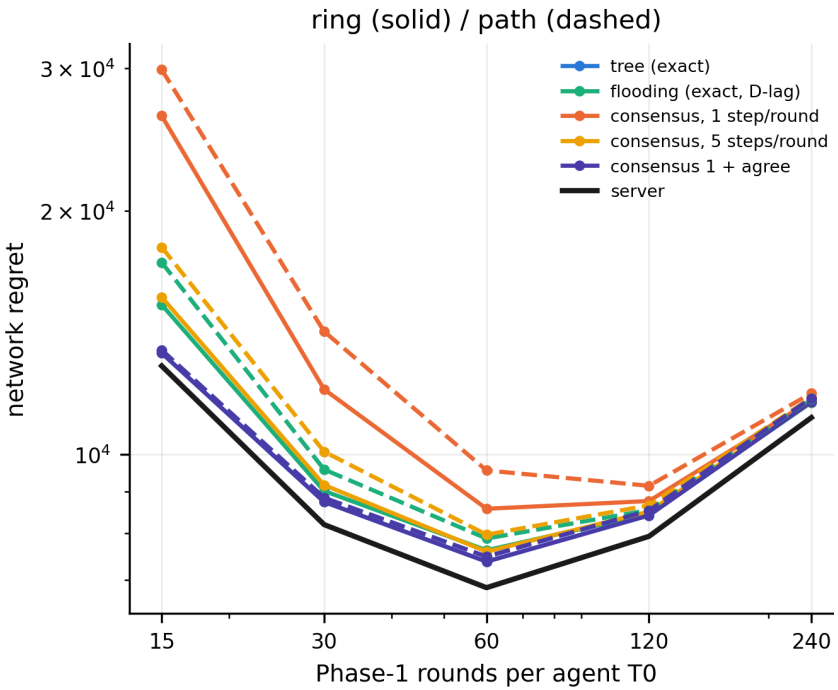
Finding 4: communication is all in Phase 2

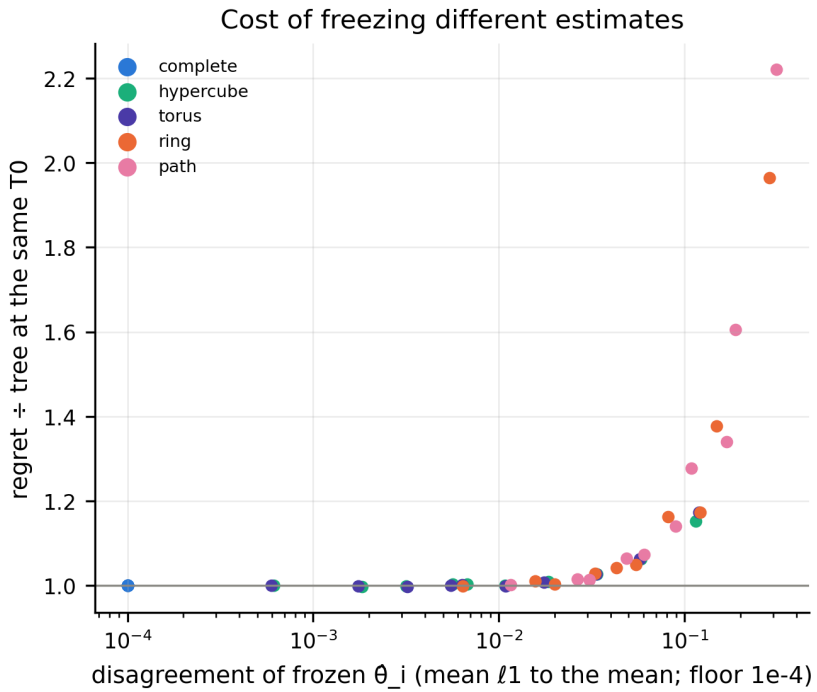




- Phase 1 is **0.1–1.6%** of all traffic ($d = 5 \ldots 200$), and shrinks with NT; compressing it saves $\leq 0.4\%$ of bits.
- Event-triggered sync: 61→213 rounds as NT grows $32\times$ (per-round sync: 1k→40k), at the same regret.

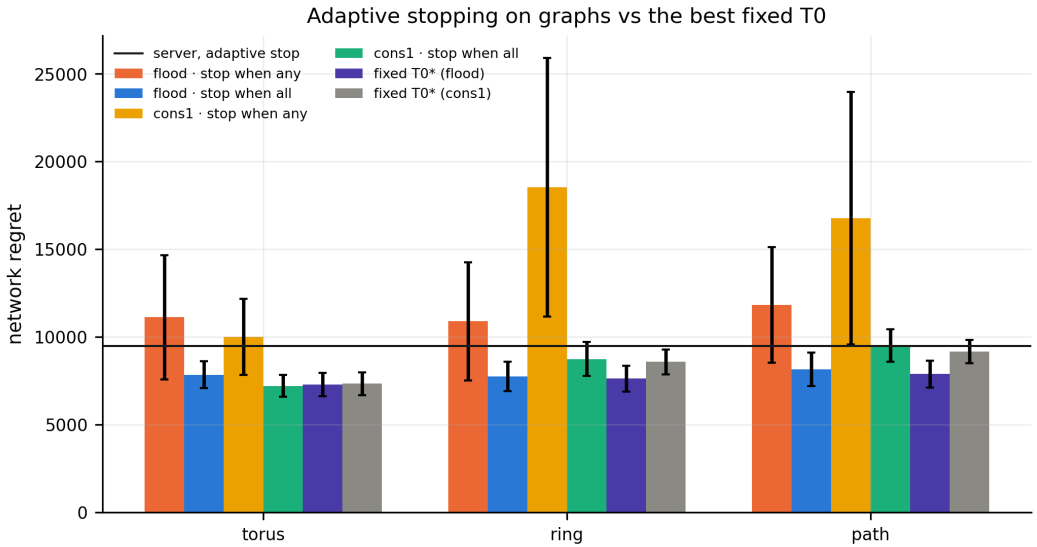
Finding 5: agents must agree, not explore longer





- The best exploration length T_0^* barely moves with the graph.
- The cost is agents freezing *different* $\hat{\theta}_i$: consensus costs $\times 1.25$ (ring), $\times 1.34$ (path).
- **One agreement pass** at the freeze removes it, for $\sim$2.8k extra scalars.

Finding 5: adaptive stopping on graphs fails



- "Stop when *any* agent is stable": **1.4–2.2**$\times$ regret, high variance.
- "Stop when *all* are": close to fixed T_0, but needs $\geq D$ extra rounds to detect (not modelled).
- Even the server's adaptive stop is worse than a fixed T_0.

Now answered (Prop 34)

Apply the server's rule to flooded pooled sums: exact agreement, Λ rounds later.

Summary of findings

Claim	Evidence
Collaboration pays without a server	regret $\propto N^{0.44\text{–}0.55}$ vs $N^{0.98}$ alone (E19)
Relaying: same regret, far less talk	$\times 1.03$ of server at its communication; mixing needs $\sim 2000\times$ more scalars (E15, E20)
Robust to failing links	$+27\%$ at 90% link failures (E18)
Server $=$ one agent at NT	within 0–10%; proved lower $+$ upper bound (E22)
No server costs little	$+1.5\%$ to $+9\%$, nearly flat in T (E23)
Graph cost has the proved shape	epochs: excess $/NDK = 0.86\text{–}1.15$; relays: excess $\propto$ farness; flat in T (E31, E36)
Provable relay $=$ the relay we ran	at $\gamma = 0.5$ same excess and scalars; at $\gamma = 1/(2N)$ lower excess, 7–8$\times$ more scalars (E36)
Dependence on K	grows like $K^{0.4}$–$K^{0.5}$; bounds are linear in K (E36)
Communication is Phase 2	Phase 1 $< 2\%$; logarithmic sync rounds (E22, E24)
Agreement is what matters	disagreement $\times 1.25\text{–}1.34$, fixed by one pass (E28); stopping with agreement (E33)
Delivery time on time-varying graphs	$B(N-1)$ is tight; benign schedules far below (E32)
Prior algorithms on our problem	best on non-monotone links, gap grows with T; monotone methods win on monotone links (E34)
Private Phase 1	provable but costly at our scale (E35)

Baselines for the serverless setting

1. Reference baselines (as in DLUCB, Coop-KernelUCB)

No communication (independent ZoomSIB) and centralised (server) ✓ done

2. Prior papers' protocols, run with ZoomSIB

Running consensus (Landgren et al.), gossip (DDUCB), accelerated consensus (DLUCB), message passing (Coop-Kernel), push-sum ✓ done

3. Prior papers' algorithms, run on our problems

- DLUCB-style (LinUCB), Dec-ESTOR, Dec-GSTOR, Coop-Kernel-style (IGP-UCB), each with *ideal* relaying of all raw data ✓ done (E34)

Status and next steps (Paper A)

Work	Deliverable	Status
Server theorem: lower + upper bound	`docs/proofs/server.pdf`	✓
Relay theorem (synchronised epochs)	Thm 17, Cor 18	✓
Communication, bounded memory	Thm 29, Prop 30	✓
Directed / delayed / time-varying / failing links	Lemma 31, Cor 32; E32	✓
Agreement; stopping with agreement	Thm 33, Prop 34; E28, E33	✓
Proved vs run protocol; trigger cost	E31	✓
Prior algorithms on our problem	E34	✓
Delay term: horizon-free, matches E31 shape	Thm 22; E31, E36	✓
A provable relay as good as the one we ran	Thm 27; E36	✓
Graph lower bound $\Omega(N \min(T, D)^{1/3})$	Thm 7, Cor 8	✓
Upper KF' vs lower $ND^{1/3}$: close the gap	open problem	❍
Privacy	Thm 35: Phase 1 only, costly (E35)	partly
Related-work section; write-up	AISTATS-style draft	❍

Paper A: the first single-index bandit over a network: optimal with a server, and what the graph costs without one.